

Riyad Babayev

EDUCATION

Bachelor of Science in **Aerospace Engineering & Computer Science**
University of Illinois at Urbana-Champaign (UIUC)
Honors: *James Scholar (GPA: 3.6/4.0)*
Relevant Courses taken: Engineering Materials & Structures, CAD, Flight Mechanics, Compressible/Incompressible Flow, Numerical Methods, Control Systems, Propulsion, Robotics, Aerodynamics, UAV Design, Senior Design

Expected graduation date: May 2026

FEATURED PROJECTS

- Next-Generation Carrier Strike Fighter (AIAA Design Competition)
- Led propulsion and landing gear design for a **Mach 2.0** single-engine carrier strike fighter, targeting FA-18E/F replacement with IOC 2035
 - Performed **compressible flow and Fanno flow analysis** to size a diverterless supersonic inlet (DSI) optimized for Mach 2.0 at 30,000 ft
 - Conducted engine cycle trade study for a modified **F110 turbofan** (BPR reduction from 0.76 to 0.70), achieving a ~7% efficiency gain using **GasTurb 14**
 - Authored the propulsion and inlet sections of a formal AIAA-standard technical report in LaTeX, documenting inlet sizing, thrust analysis, and airworthiness compliance per MIL-HDBK-516C
- Racing Quadrotor Drone
- Developed and simulated real-time control algorithms (LQR + feedforward) in **Python** for high-speed drone racing
 - Leveraged **NumPy, Matplotlib, and SymPy** for numerical computation, modeling, and data visualization
 - Designed data-driven tuning scripts to ensure rapid response stability under maneuvers, reducing response lag by 20%

WORK EXPERIENCE

- INTERNSHIP: **Audubon Companies (Engineering Consulting Company), Houston, USA**
Field of work: **Mechanical Engineering & Software Automation**
Summer 2025 & 2024
- Designed and 3D printed a *24" Ethane Trap Skid* for client demonstration and integration
 - Implemented automated data formatting pipelines to eliminate repetitive manual entry, reducing processing time by 60%
 - Utilized **AutoCAD** for modeling and drafting mechanical components and crafting the P&IDs
 - Integrated document management systems using conditional programming logic to interconnect data between engineering, procurement, and compliance teams
 - Drafted mechanical engineering lists and drawings, referencing P&IDs and Pipe Specifications
 - Engineered pipe and valve specifications for a CO₂ service project
 - Pursued wall thickness calculations along with stiffness using **Nozzle PRO**
 - Conducted pipe stress analysis using **CAESAR CX II** in accordance with ASME standards
- Field of work: **Project Management**
- Supplied clients with weekly updates, reports, RFIs, ensuring information was current and accurate
 - Controlled schedules and deliverables for activities for \$52.7M refinery development project
 - Managed a blast impact study for refinery reactors in compliance with laws, addressing proximity to two civilian buildings
 - Collaborated with 7 vendors to meet client requirements, ensuring timely delivery and integration of critical components
- INTERNSHIP: **AZAL (Azerbaijani Airlines), Azerbaijan**
June – August 2023
- Consistent daily checks and monitoring of technical parameters (Diagnostics) of aircraft
 - Fulfilling requirements on MOE and CAME (Continuing Airworthiness Management Exposition)
 - Analyzed over 107 repetitive defects in the fleet and monitored them to prevent future complication
 - Pursuing on-site aircraft inspection and reporting defects and dents

STUDENT ORGANIZATIONS INVOLVEMENT

- ORGANIZATION: **ASAUIUC (Azerbaijan Student Association in UIUC)**
POSITION: **President**
August 2024 - Current
- Founded** the first registered Azerbaijani Student Organization at UIUC, aiding in cultural exchange and community building
 - Expanded membership from 5 to 25 members in one semester by organizing meeting, cultural events, and guest lectures
 - Secured \$3,500 in event funding through successful Student Organization Resource Fee (SORF) grant applications
- ORGANIZATION: **GHOST Electric Motorcycles**
POSITION: **Mechanical Engineer**
October 2024 – January 2025
- Transitioned the team's design workflow to AutoCAD, improving design precision and reducing drafting errors by 9%, leading to faster prototyping and iteration
 - Led a team of 18 members to design and improve an electric motorcycle, resulting in a 20% increase in efficiency
 - Conducted tests and data analysis on 5+ motor configurations, optimizing torque output and increasing top speed

SKILLS

- Software
- Programming:** Java, Python, C++, AVL
 - CAD Software:** NX Siemens, AutoCAD, SolidWorks, Navisworks
 - FEA & Analysis tools:** Cura, ANSYS, ABAQUS, Nozzle PRO, CAESAR CX II, GasTurb 14
 - Microsoft Office:** Word, Excel, PowerPoint | Bluebeam Revu
- Languages
- English, Azerbaijani, Russian, Turkish